talk 4: Elliptic curves over local fields

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Introduction

This is my note for Talk 4. The main references are Andrew Snowden's lectures and Silverman's textbook. The schedule:

- Basics: Elliptic Curves over a general scheme, Weierstrass equations
- Reduction types of elliptic curves over local fields
- Torsions, formal groups
- Good reduction, criterion of Néron-Ogg-Safarevic

Basics(See Mazur's book Arithmetic Moduli of Elliptic Curves)

Definition 0.1. Let S be any scheme. An elliptic curve over S is a proper smooth family of curves $f : E \to S$ whose fibers are geometrically connected of genus 1 together with a section "O" : $S \to E$.

Definition 0.2 (Group Laws)**.** For any $T \to S$, $P + Q = R$ in $E(T)$ iff there exists some line bundle $\mathcal{L}_0$ on T such that $\mathcal{L}(P) \otimes \mathcal{L}(Q) \otimes \mathcal{L}^{-1}(O) \cong \mathcal{L}(R) \otimes f_T^*(\mathcal{L}_0)$

Definition 0.3 (Invariant Differentials)**.** Grothendieck-Serre duality gives $R^1 f_* \Omega^1_{E/S} \cong \mathcal{O}_S$. Therefore $\omega_{E/S} = f_* \Omega^1_{E/S}$ is an invertible sheaf on S, by Riemann-Roch. Zariski locally on S, we choose a $\mathcal{O}_S$-basis ω of $\omega_{E/S}$. ω is a nowhere vanishing 1-form on E. It is translation invariant: $\lambda_g^* \omega$ differs from ω by something $\in \mathbb{G}_m$ hence defines a morphism $E \to \mathbb{G}_{m,S}$ which is constant.

Definition 0.4 (Weierstrass Equations)**.** $f_* \mathcal{L}^n(O)$ is locally free of rank n, by Riemann-Roch.

- $f_* \mathcal{L}^2(O)$: free on $1, x$
- $f_* \mathcal{L}^3(O)$: free on $1, x, y$
- $f_* \mathcal{L}^4(O)$: free on $1, x, y, x^2$
- $f_* \mathcal{L}^5(O)$: free on $1, x, y, xy, x^2$

Let T be a local uniformizing parameter for $\hat{\mathcal{O}}_{O,E}$. Then locally

$$\omega = (1 + T \cdot ...)dT$$
$$x = \frac{1}{T^2}(1 + ...)$$
$$y = \frac{1}{T^3}(1 + ...)$$

. $y^2 - x^3 \in f_*\mathcal{L}^5(O)$. We hence find the Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Theorem 0.1. *The multiplication-by-N map $[N] : E \longrightarrow E$ is finite flat.*

- *Properness: Since morphisms between proper schemes are automatically proper, we only need to show $[N]$ has finite fibers. This is easy.*
- *Flatness: Finite morphisms between regular schemes of the same dimension are flat. The total space of E over S is regular.*

Reduction Types of ECs over local fields

Fix some notations first. We work over a complete discrete valuation ring R with quotient field K.

- valuation $v : R \to \mathbb{Z} \cup \{\infty\}$
- maximal ideal $\mathfrak{m}$.
- uniformizer π
- residue field k

We start with an elliptic curve E over K. We want to find a minimal Weierstrass equation. This is characterized by the minimality of $v(\Delta)$, equivalently $v(\Delta) < 12$ or $v(c_4) < 4$ or $v(c_6) < 6$ where Δ is the discriminant of Weierstrass equation, and $c_4 = a, c_6 = b$ if $\mathrm{char}(k) > 5$. We adopt the following notations, given a model $\mathcal{E}$ defined by a minimal Weierstrass equation:

- $\overline{E}$: the special fiber of $\mathcal{E}$
- $\overline{E}_{sm}$: smooth locus of $\overline{E}$, a group isomorphic to an elliptic curve, $\mathbb{G}_a$ or $\mathbb{G}_m$
- $E_0(K)$: the set of $E(K)$ reducing to a point in $\overline{E}_{sm}$, the reduction map defined by dialating the homogeneous coodinates such that some entry is in R^*.
- $E_1(K)$: kernel of reduction map (isomorphic to the formal group $\hat{E}(\mathfrak{m})$)

Example 0.1.

$$\begin{aligned} E : y^2 + xy + y &= x^3 + x^2 + 22x - 9 \\ \Delta &= -2^{15}5^2 \\ c_4 &= -5 \cdot 211 \end{aligned}$$

This is minimal

Proposition 0.1.

- *(a) every EC has a minimal Weierstrass equation.*
- *(b) a minimal equation is unique up to a change of coordinates*

$$\begin{aligned} x &= u^2x' + r \\ y &= u^3y' + u^2sx + t \\ u &\in R^*, r, s, t \in R \end{aligned}$$

, and further if $\mathrm{char}(k) > 5$, we force the equation to be $y^2 = x^3 + ax + b$, it is unique up to $x = u^2x', y = u^3y'$

- *the invariant differential* $\omega = \frac{dx}{2y+a_1x+a_3}$ *is unique up to something in* R^*.
- *any minimal Weierstrass equation is produced this way.*

Theorem 0.2. *there is an exact sequence*

$$0 \longrightarrow E_1(K) \longrightarrow E_0(K) \longrightarrow \overline{E}_{ns}(k) \longrightarrow 0$$

Proof. Surjectivity: Given $\overline{P} \in \overline{E}_{ns}(k)$, either $\frac{\partial f}{\partial x}(\overline{P})$ or $\frac{\partial f}{\partial y}(\overline{P})$ is nonzero, we say the latter. Then an easy application of Hensel's lemma imples the result.
The subtle part is that $E_0(K)$ is a group. We leave this, see Silverman's book, exercise 7.15.

□

Definition 0.5. Reduction Types:

- good reduction: $\overline{E}$ is nonsingular.
- multiplicative reduction: $\overline{E}$ has a node, and $\overline{E}_{sm} \cong \mathbb{G}_m$
- split multiplicative reduction: slopes of 2 tangents of the node are $\in k$
- additive reduction: $\overline{E}$ has a cusp, and $\overline{E}_{sm} \cong \mathbb{G}_a$

The latter three are called bad reduction. The first three are called semistable reduction.

Proposition 0.2. *SUppose we are given a model* $\mathcal{E}$ *of* E *defined by a minimal Weierstrass equation*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

then

- *(a) good reduction* $\iff v(\Delta) = 0$
- *(b) multiplicative reduction* $\iff v(\Delta) > 0, v(c_4) = 0$
- *additive reduction* $\iff v(\Delta) > 0, v(c_4) > 0$

Example 0.2.

$$E_1 : y^2 = x^3 + px^2 + 1$$
$$E_2 : y^2 = x^3 + x^2 + p$$
$$E_3 : y^2 = x^3 + p$$

the third one has additive but potentially good reduction.

Theorem 0.3 (semistable reduction theorem)**.**

- *(a) If* K'/K *is unramified, then* E_K *and* $E_{K'}$ *has the same reduction type*
- *(b) If* K'/K *is finite, and* E *has semistable reduction* $\implies E_{K'}$ *has semistable reduction*
- *There exists a finite extension* K'/K *such that* E *has good or split multiplicative reduction over* K'.

Proof. For simplicity assume $\mathrm{char}(k) > 5$. The minimal Weierstrass equation over K is given by $y^2 = x^3+ax+b$. If the minimal Weierstrass equation over K' is given by the coordinate change $x \mapsto u'^2x', y \mapsto u'^3y'$, we can choose $u \in R$ with $u/u' \in R'^*$ and make the coordinate change $x \mapsto u^2x', y \mapsto u^3y'$ then $v'(u^{-12}\Delta) = v'(u'^{-12}\Delta)$. But there is no space for u so the original equation is minimal, hence $E_{K'}$ has the same reduction type.
For the second statement, still take a minimal equation over K first. Suppose $x \mapsto u^2x', y \mapsto u^3y'$ transforms the original equation into minimal over K'. By the transformation formulas for c_4, Δ we have $v'(u^{-12}\Delta) \geq 0, v'(u^{-4}c_4) \geq 0$. So $v'(u) = v(u) = 0$.
For the third statement, we exploit the fact that every elliptic curve is isomorphic to a elliptic curve given by a Legendre form $y^2 = x(x-1)(x-\lambda)$ over $\overline{K}$. □

Theorem 0.4. ***Potentially good reduction*** $\iff j(E) \in R$

Proof. Take some extension to obtain the Legendre form

$$y^2 = x(x-1)(x-\lambda), \lambda \neq 0, 1$$

and the j-invariant $j = j(E)$ satisfies

$$256(1-\lambda(1-\lambda))^3 - j\lambda^2(1-\lambda)^2 = 0$$

$j \in R$ imples $\lambda \in R$ hence good reduction otherwise $v(j) < 0$, contradiction.
Conversely, if E has potentially good reduction, $v'(\Delta) = 0$ for some $v'|v$. Then $j(E) = \frac{(c_4')^3}{\Delta'} \in R'$. So $j(E) \in R$ □

Torsion, formal groups

To summarize, we start from the exact sequence

$$0 \longrightarrow E_1(K) \longrightarrow E_0(K) \longrightarrow \overline{E}_{ns}(k) \longrightarrow 0$$

and analyzed possiblities of $\overline{E}_{ns}(k)$. We also need to cite a fact: $E(K)/E_0(K)$ is finite. $|E/E_0|$ equals to $-v(j)$ when E has split multiplicative reduction and ≤ 4 in other cases. We still need to know about E_1.

Definition 0.6. Consider a group scheme (E, m) over $\mathrm{Spec}(R)$, in our case an elliptic curve. $m^* : \mathcal{O}_E \longrightarrow \mathcal{O}_E \otimes \mathcal{O}_E$ induces $\hat{m}^* : \hat{\mathcal{O}}_{O,E} \to \hat{\mathcal{O}}_{O,E} \otimes \hat{\mathcal{O}}_{O,E}$, where $\hat{\mathcal{O}}_{O,E} = \varprojlim_n \mathcal{O}_E/\mathcal{I}_O^n$. In our case $\hat{\mathcal{O}}_{O,E} = R[[z]]$. So this map is given by a power series in two variables $F(u, v)$. Call this formal group $\hat{E}$

Proposition 0.3.

$$\begin{aligned} F(x,y) &= x + y + (\deg \geq 2) \\ F(x, F(y,z)) &= F(F(x,y), z) \\ F(x,y) &= F(y,x) \\ \exists! i(T) \in R[[T]], F(T, i(T)) &= 0 \\ F(X,0) = X, F(0,Y) &= Y \end{aligned}$$

How to find the group law algorithm defined by an elliptic curve? Let $z = -\frac{x}{y}, w = \frac{1}{y}$. Then z is a uniformizer at O. $w = z^3 + a_1zw + a_2z^2w + a_3w^2 + a_4zw^2 + a_6w^3$. Call the right hand side $f(z, w)$.

Iterate w by this equation into the equation repeatedly yields some power series $w(z)$ in z, satisfying $f(z, w(z)) = w(z)$. Let $\lambda = \lambda(z_1, z_2) = \frac{w(z_2)-w(z_1)}{z_2-z_1}$. Then

$$\begin{aligned} w &= \lambda z + w(z_1) - \lambda z_1 \\ f(z, w) &= 0 \\ w(z_1) - \lambda z_1 &= \nu \end{aligned}$$

yields

$$z_3 = -z_1 - z_2 - \frac{a_1\lambda + a_3\lambda^2 + a_2\nu + 2a_4\lambda\nu + 3a_6\lambda^2\nu}{1 + a_2\lambda + a_4\lambda^2 + a_6\lambda^3}$$

The assignment $z \mapsto (\frac{z}{w(z)}, -\frac{1}{w(z)})$ defines an isomorphism $E_K(\mathfrak{m}) \cong \hat{E}(\mathfrak{m})$.

Proposition 0.4.

$$\frac{\hat{E}(\mathfrak{m}^n)}{\hat{E}(\mathfrak{m}^{n+1})} \cong \mathfrak{m}^n/\mathfrak{m}^{n+1} \cong k.$$

Hence for $m \nmid \mathrm{char}(k), x \in \hat{E}(\mathfrak{m}), mx \neq 0$.

Corollary 0.1. *If E has good reduction,$m \nmid \mathrm{char}(k)$, $E(K)[m] \to \overline{E}(k)$ is injective.*

Criterion of Néron-Ogg-Safarevic

Theorem 0.5.

- *(a) E has good reduction $\iff I_K \curvearrowright T_\ell(E)$ trivially*
- *(b) Semistable reduction $\iff I_K \curvearrowright T_\ell(E)$ unipotently*

Proof. (a):
($\Longleftarrow$):
K^{ur} is a discrete valuation field. So $E(K^{ur})/E_0(K^{ur})$ is finite, say of cardinality d. $T_\ell(E)$ has trivial I_K-action iff $I_K \curvearrowright E(\overline{K})[\ell^n]$ trivially for all n. Now $E(\overline{K})[\ell^n] = E(K^{ur})[\ell^n]$ and this group has cardinality ℓ^{2n}. So $E_0(K^{ur})[\ell^n]$ has cardinality at least ℓ^{2n}/d. But $E_0(K^{ur})[\ell^n] \to \overline{E}_{sm}(\overline{k})[l^n]$ is injective, so $|\overline{E}_{sm}(\overline{k})[\ell^n]| \geq \ell^{2n}/d$. Hence $\overline{E}_{sm}$ cannot be $\mathbb{G}_a$ or $\mathbb{G}_m$.
($\Longrightarrow$):
We have $E(\overline{K})[\ell^n] \to E(\overline{k})[\ell^n]$ is injective. This follows from $E_0(\overline{K}) = E(\overline{K})$ since E has good reduction, and that $E_{0,\ell-tors} \to \overline{E}_{sm,\ell-tors}$ is injective. This map is also Galois-equivariant. The latter is unramified so the former is also.
(b):
($\Longleftarrow$):
I_K fixes some vector $v \in T_\ell$ so $|\overline{E}_{sm}(\overline{k})[\ell^n]| \geq |E(K^{ur})[\ell^n]| \geq \ell^n$. This is impossible for additive reduction.
($\Longrightarrow$):
Let $\mathcal{E}$ be the given semistable model. $\mathcal{E}_{sm}[\ell^n]$ is a flat group scheme. Let G be the schematic closure in $\mathcal{E}_{sm}[\ell^n]$ of $\overline{K}$-points which extends to $\overline{R}$-points. So G finite flat and $G_k = \overline{E}_{sm}[\ell^n]$. Now G has ℓ-power order hence G étale. So $G(K^{ur}) = \overline{E}_{sm}[\ell^n](\overline{k})$, which contains $\mathbb{Z}/\ell^n\mathbb{Z}$ by semistablity. So $T_\ell(E)$ has a fixed vector. But $\det(T_\ell) = \chi_{cyc,\ell}$ which is trivial on I_K. □